Brain Tumor Detection Using Deep Learning

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***Abstract*—** Brain tumor detection is a vital task in the medical field, where early diagnosis significantly enhances patient survival chances. This study implements a Convolutional Neural Network (CNN) to detect brain tumors from MRI images. Utilizing a structured dataset and advanced preprocessing, the model achieves high accuracy in classification. The proposed deep learning model provides a fast, reliable, and automated diagnostic tool that can assist healthcare professionals in clinical settings.

***Index Terms*—** Brain Tumor, Deep Learning, Convolutional Neural Network, Image Classification, Medical Imaging

I. Introduction

Brain tumors are abnormal growths of cells in the brain and can be life-threatening. Traditional diagnostic processes often rely on manual MRI scans, which are time-consuming and prone to human error. The rise of Artificial Intelligence (AI), particularly Deep Learning, has shown promising results in automating and enhancing the accuracy of medical diagnostics. This paper presents a CNN-based approach to automatically classify MRI images as tumor or non-tumor.

II. Materials and Methods

A Sequential CNN model was implemented with the following architecture:

- Input Layer: Accepts 224×224 RGB images.
- Convolutional Layers: Three Conv2D layers with ReLU activation and increasing filter sizes (32, 64, 128).
- Pooling Layers: MaxPooling2D after each Conv2D to reduce dimensionality.
- Flatten Layer: Converts the output into a 1D vector.
- Fully Connected Layers: Dense(128) with ReLU, Dropout(0.5) for regularization.
- Output Layer: Dense(1) with sigmoid activation for binary classification.

III. Dataset Description

The dataset used comprises MRI images categorized into two classes: tumor and no tumor. Images are stored in a structured directory with subfolders representing the two classes. Preprocessing steps included image resizing to (224×224), normalization (dividing pixel values by 255), and label encoding.

IV. Training Procedure

The dataset was split into 80% training and 20% validation sets using stratified sampling to maintain class balance. The model was compiled with:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Metrics: Accuracy.

Results

The model demonstrated an accuracy of 97.3% on the validation dataset. Accuracy and loss graphs indicated proper convergence without overfitting. The model's ability to generalize well on unseen data suggests its practical applicability in real-world medical environments.

Refer to **Fig.1**

Discussion

The experimental outcomes of our CNN-based brain tumor detection model demonstrate promising performance in classifying MRI brain scans into tumor and non-tumor categories. With an accuracy exceeding 96%, the model effectively learns spatial hierarchies and patterns indicative of tumorous growth. The architecture—comprising multiple convolutional and pooling layers followed by dense layers—proves to be efficient in extracting features even from relatively small datasets.

However, there are certain limitations. While the model generalizes well on validation data, its performance may degrade when exposed to images from different MRI machines or varying contrast conditions not seen during training. Additionally, class imbalance (often skewed towards non-tumor images in public datasets) can bias predictions. Techniques like data augmentation and oversampling were partially used to counter this, but further refinement is possible

Conclusion

In this study, we developed and evaluated a Convolutional Neural Network for brain tumor detection using MRI images. The model achieved an accuracy of over 96%, demonstrating its capability to accurately distinguish between tumor and non-tumor brain scans. The results underline the potential of deep learning in enhancing diagnostic accuracy, particularly in early-stage tumor detection. .

Despite its effectiveness, the system's performance can be further improved by incorporating more diverse datasets, applying advanced image pre-processing techniques, and experimenting with more sophisticated architectures. The findings of this study contribute to the growing field of AI-powered medical imaging and serve as a foundation for future real-time and mobile-based diagnostic tools

Fig.1

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